

Vansh

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Summary

Professional Full-Stack and Machine Learning Engineer with 2 years of experience in scalable web development and 1 year of experience building high-accuracy ML models. Demonstrated expertise in deploying real-time AI solutions, optimizing multi-source data pipelines, and architecting robust systems using the MERN stack, Python, and AWS to drive measurable performance improvements.

Education

B.Tech in Computer Science, VIT Bhopal

Sept 2023 – Aug 2027

Class XII, Floradale School, Ganganagar, Rajasthan

84.8%

Class X, D.A.V. Public School, Suratgarh, Rajasthan

95.33%

Skills

Languages: Java, Python, C++, JavaScript

Web Technologies & Libraries: React, Node.js, Express.js, Flask, FastAPI, HTML/CSS, Tailwind CSS, Material-UI, Socket.io, JWT

Machine Learning & AI: TensorFlow, PyTorch, XGBoost, Pandas, NumPy, Scikit-learn, Large Language Models (LLMs), Prompt Engineering

Databases: MongoDB, MySQL, PostgreSQL, Oracle Database

Cloud & DevOps: AWS, Oracle Cloud Infrastructure (OCI), Docker, CI/CD

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), REST APIs

Experience

Web Development Intern — KL LAB

May 2025 – Jul 2025 | Remote

- Implemented secure and scalable REST APIs using Node.js and Express.js, implementing JWT-based authentication that protected 20+ backend endpoints and reduced unauthorized access attempts by 30%.
- Architected optimized MongoDB schemas via Mongoose, accelerating data retrieval efficiency for user, blog, and comment workflows by 40% and reducing query latency to under 50ms
- Deployed backend systems to cloud infrastructure, collaborating with a 4-person frontend team to achieve seamless full-stack integration and maintain 99.9% uptime post-launch.

Technical Team Core Member — AWS Club, VIT Bhopal

Oct 2024 – Jul 2025 | Bhopal, MP

- Spearheaded 5+ technical workshops on AWS fundamentals for 100+ members, driving a 20% increase in club engagement and leading team-based cloud architecture projects.

Projects

Real-Time Network Intrusion Detection System

GitHub

Live Demo

Python, XGBoost, React, WebSockets, Docker

- Developed a high-performance XGBoost Binary Classifier trained on 421,959 cleansed network flows from the CIC-DDoS2019 dataset, achieving 99.85% accuracy in distinguishing malicious from benign traffic.
- Streamlined a custom Python data pipeline using Scapy to extract 15 distinct statistical features from live network flows for real-time edge inference.
- Architected a zero-latency SOC (Security Operations Center) dashboard using React and WebSockets, pushing threat alerts and live packet graphs to the client in under 2 seconds.
- Orchestrated a [4]-container microservices environment utilizing Docker Compose, isolating the Node.js, React, and Python engines to reduce local deployment time by [80]% (under [2] minutes).

SureBet

GitHub

Live Demo

React, Node.js, Express.js, MongoDB, Socket.io

- Architected a real-time sports arbitrage discovery platform scanning 15+ sportsbooks, featuring instant data updates via WebSockets to deliver high-value betting opportunities with sub-200ms latency.
- Programmed advanced filtering and sorting algorithms across 10+ data points, enabling users to isolate high-yield opportunities 50% faster.
- Pioneered a full-stack architecture using Node.js, Express, and MongoDB, facilitating scalable backend logic and providing secure data handling for over 100+ data points and improved data retrieval speeds 2x.
- Implemented a virtualized data table within the dashboard with sortable columns and live stake breakdowns, utilizing virtual scrolling to render 5,000+ data rows without frame drops, improving client-side load speeds by 60%.

Learnstitution

GitHub

React, Flask, Python, Node.js, Google Gen AI API

- Formulated a full-stack legal education platform to simplify complex Indian laws, improving accessibility for 500+ simulated users and decreasing the average time to find relevant legal statutes by 50%.
- Integrated an AI-powered chatbot via Flask and the Google Generative AI API, delivering real-time answers to 100+ diverse legal queries and reducing manual search time by 70%.
- Engineered interactive, game-based learning modules using React, increasing average user session duration by 35% and improving user retention rates by 25%.

Certifications

- Oracle Cloud Infrastructure 2025 Certified DevOps Professional
- Oracle Cloud Database Services 2025 Certified Professional
- MATLAB
- AWS Cloud Club - Machine Learning Camper
- The Bits and Bytes of Computer Networking - Google